GOVERNMENT OF WESTERN AUSTRALIA

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Department of Mines, Industry Regulation and Safety

242634.1.pdf *Geochronology Record 1938*

242634.1: hornblende–biotite metagranodiorite, Koolanooka Hills

(Walganna Suite, Bald Rock Supersuite; Youanmi Terrane, Yilgarn Craton)

SS Romano, MTD Wingate, Y LU, IOH Fielding and TJ Ivanic

Location and sampling

PERENJORI SH50-06, PERENJORI 2139 Zone 50, MGA 423759E 6772980N
Warox Site TJIKNK190050 Sampled on 3 May 2019

This sample was collected from bouldery outcrop north of the Koolanooka Hills, about 9.7 km southwest of Sandalwood Bore, 3.7 km south-southeast of Charley Well, and 1.9 km northeast of the Koolanooka open pit mine.

Geological context

The unit sampled is a metagranodiorite of the Walganna Suite (Van Kranendonk et al., 2013; Zibra, 2021: ENS report Walganna Suite, viewed 31 May 2023) (Fig. 1), within the northern Koolanooka greenstone belt (Baxter and Lipple, 1985). The 2650–2600 Ma Walganna Suite consists mainly of equigranular, biotite±muscovite monzogranite, locally K-feldspar porphyritic, that commonly exhibits a magmatic foliation. As one of the youngest units in the Youanmi Terrane, monzogranite of the widespread Walganna Suite is intruded into older granitic rocks and greenstones of the Murchison and Southern Cross Supergroups (Zibra, 2021: ENS report Walganna Suite, viewed 31 May 2023; Ivanic et al., 2022). The Koolanooka greenstone belt consists mainly of greenschist to amphibolite facies volcanic, sedimentary, and ultramafic to felsic intrusive rocks, and banded iron formation (Baxter and Lipple, 1985). A felsic metavolcanic rock, sampled about 11.2 km to the southeast, yielded an igneous crystallization age of 2961 ± 4 Ma and an inherited zircon component dated at 2994 ± 3 Ma (GSWA 242652, Wingate et al., 2021a). A metamonzogranite collected about 22 km to the east-northeast yielded an igneous crystallization age of 2946 ± 4 Ma (GSWA 219311, Wingate et al., 2020). A metarhyolite sampled in the southern Koolanooka greenstone belt, about 26 km to the south-southeast, yielded an igneous crystallization age of 3002 ± 4 Ma (GSWA 242672, Wingate et al., 2021b).

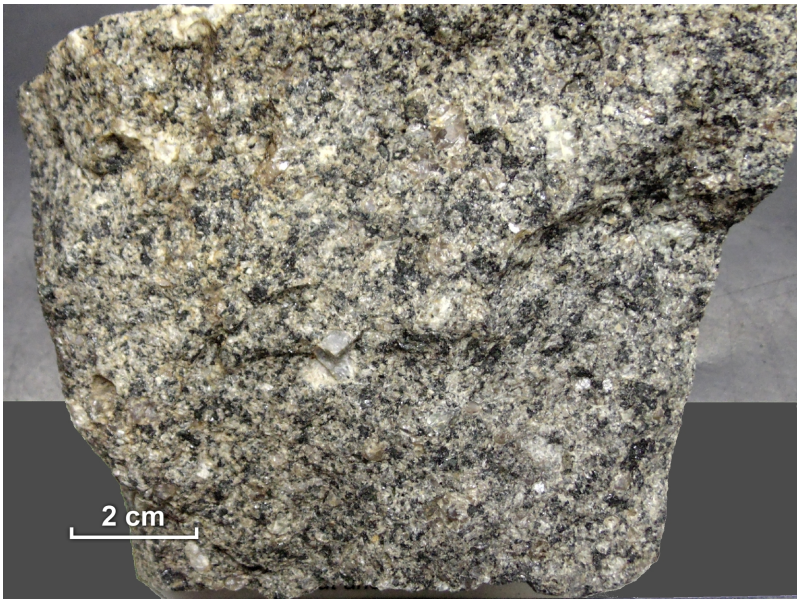

Figure 1. Hand specimen image for sample 242634: hornblende–biotite metagranodiorite, Koolanooka Hills

Petrographic description

This sample is a medium-grained, recrystallized, porphyritic hornblende–biotite metagranodiorite (Fig. 1), consisting of about 35% plagioclase, 25% quartz, 15% K-feldspar, 12–15% hornblende, 8% biotite, 3–4% opaque minerals, and accessory titanite, apatite and zircon. Plagioclase (An_{38}, andesine) is weakly sericitized, and forms subhedral to euhedral,

tabular phenocrysts up to 5 mm long, as well as subhedral grains up to 0.25 mm in size. Quartz is weakly to moderately strained, and forms anhedral phenocrysts up to 3 mm diameter, as well as equant, irregular grains up to 0.25 mm in size, together with feldspars in granoblastic aggregates. Green hornblende is anhedral, tabular, up to 2 mm long, with interior zones of actinolite and minor magnetite, which may represent altered original pyroxene subsequently rimmed by hornblende. Some hornblende contains tiny quartz inclusions. Fined-grained glomeroporphyritic aggregates of hornblende, plagioclase, biotite and magnetite are up to 8 mm in size. Yellow to brown biotite forms anhedral to subhedral laths and decussate aggregates of subhedral grains up to 0.25 mm long. Magnetite is anhedral, slightly embayed, up to 1 mm in size, and locally intergrown with or rimmed by titanite. Rhombic titanite crystals up to 0.15 mm in size, and euhedral apatite up to 0.6 mm long, occur within hornblende-rich aggregates. The groundmass contains patches of fine-grained granophyre up to 1 mm in size. The sample lacks a tectonic fabric, but is extensively recrystallized, indicating contact metamorphism to hornblende hornfels facies.

Zircon morphology

Zircons isolated from this sample are colourless to dark red–brown or opaque, and mainly euhedral. The crystals are up to 500 µm long, and mainly elongate to acicular, with aspect ratios up to 9:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, some zircons contain embayed cores, and many grains contain inclusions. A CL image of representative zircons is shown in Figure 2.

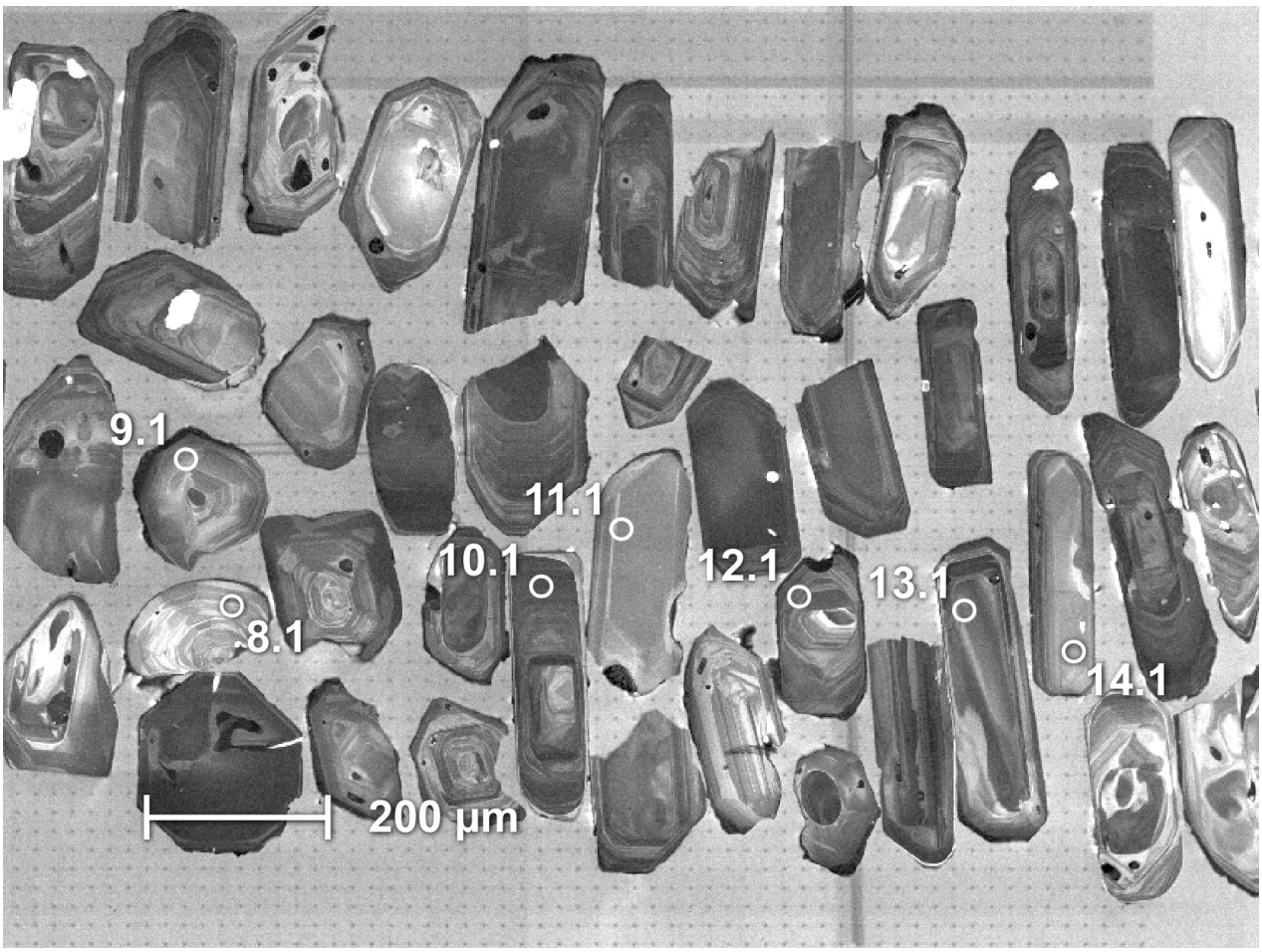

Figure 2. Cathodoluminescence image of representative zircons from sample 242634: hornblende–biotite metagranodiorite, Koolanooka Hills. Numbered circles indicate the approximate locations of analysis sites

Analytical details

This sample was analysed on 24–25 September 2020, using SHRIMP-B. Twelve analyses of the M257 standard were obtained during the session, of which 10 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.25% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1 (provided in the Links section). Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Eighteen analyses were obtained from 18 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to slightly discordant (Fig. 3), and form a single group, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 18 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2627 ± 6 Ma (MSWD = 1.2).

The date of 2627 ± 6 Ma for the 18 analyses in Group I is interpreted as the igneous crystallization age.

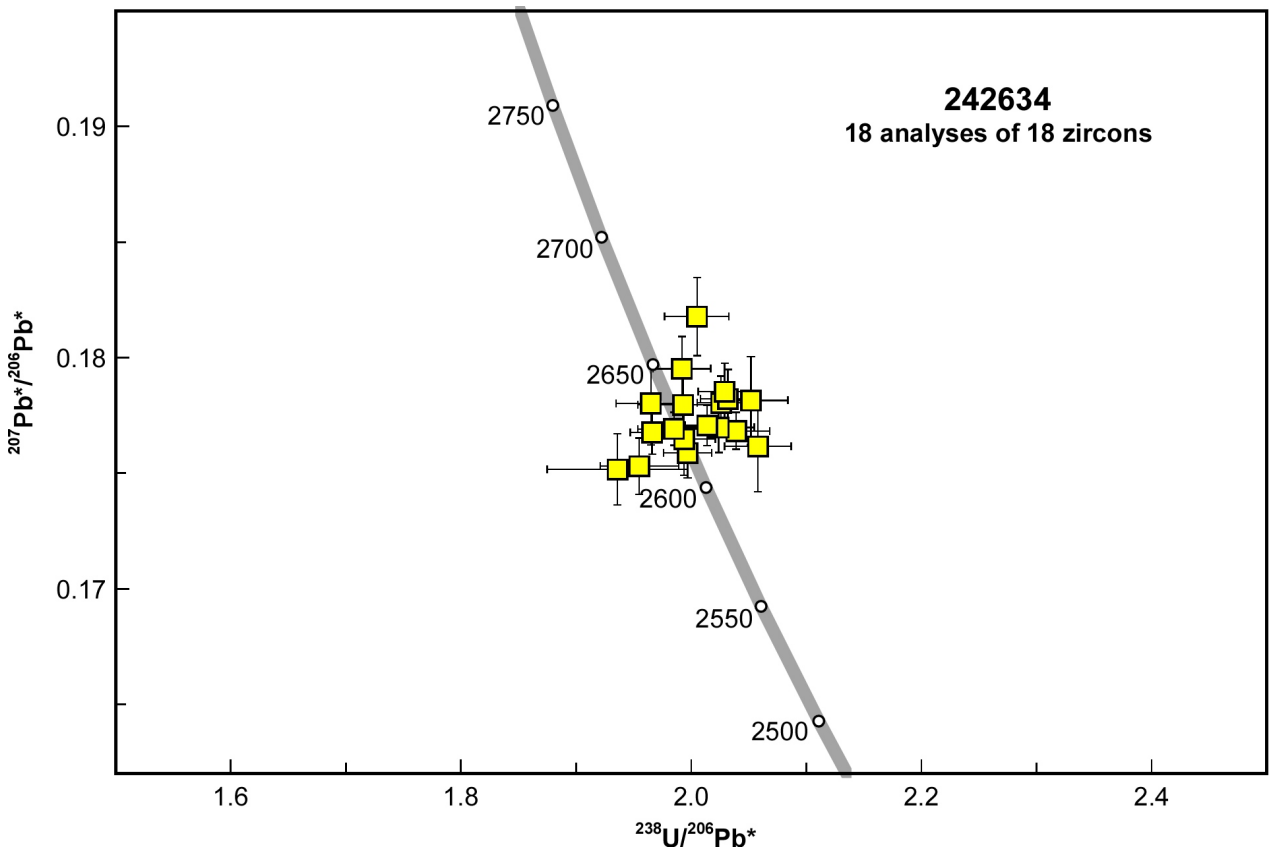

Figure 3. U–Pb analytical data for zircons from sample 242634: hornblende–biotite metagranodiorite, Koolanooka Hills. Yellow squares indicate Group I (magmatic zircons)

Links

Table 1: Ion microprobe analytical results for zircons from sample 242634: hornblende–biotite metagranodiorite, Koolanooka Hills: 242634.1.Table1.xls
SHRIMP spot images: 242634_spots.pdf
Introduction to geochronology information : Intro_2023.pdf

References

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Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

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Recommended reference for this publication

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Grid references in this publication refer to the Geocentric Datum of Australia 2020 (GDA20). All locations are quoted to at least the nearest 100 m.

Capitalized names in text refer to standard 1:100 000 map sheets, unless otherwise indicated.

WAROX is GSWA's field observation and sample database. WAROX site IDs have the format 'ABCXXXnnnnnnSS', where ABC = geologist username, XXX = project or map code, nnnnnn = 6-digit site number, and SS = optional alphabetic suffix (maximum 2 characters).

Isotope and element analyses are routinely conducted using the GeoHistory laser ablation ICP-MS and Sensitive High-Resolution Ion Microprobe (SHRIMP) facilities at the John de Laeter Centre (JdLC), Curtin University, with the financial support of the Australian Research Council and AuScope National Collaborative Research Infrastructure Strategy (NCRIS), and the Microscopy Australia laser ablation ICP-MS facility at the Centre for Microscopy, Characterisation & Analysis (CMCA), The University of Western Australia (UWA). The TESCAN Integrated Mineral Analyser (TIMA) instrument was funded by a grant from the Australian Research Council (LE140100150) and is operated by the JdLC with the support of the Geological Survey of Western Australia, UWA and Murdoch University. Mineral analyses are routinely obtained using the electron probe microanalyser (EPMA) facilities at the CMCA at UWA, at Adelaide Microscopy at the University of Adelaide and the Central Science Laboratory at the University of Tasmania.

Digital data related to WA Geology Online, including geochronology and digital geology, are available online at the Department's Data and Software Centre and may be viewed in map context at GeoVIEW.WA. For further information about this Geochronology Record, please contact geochronology@dmirs.wa.gov.au and quote the geochronology record and/or the sample number.

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